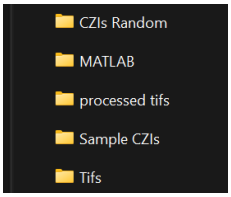
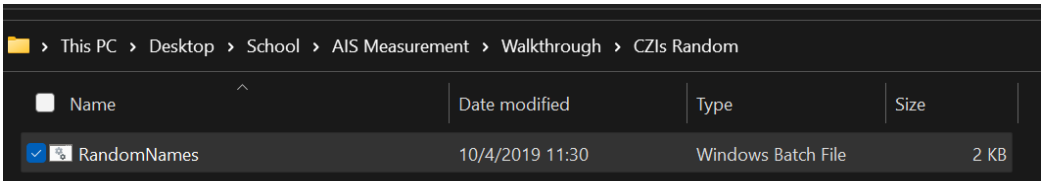
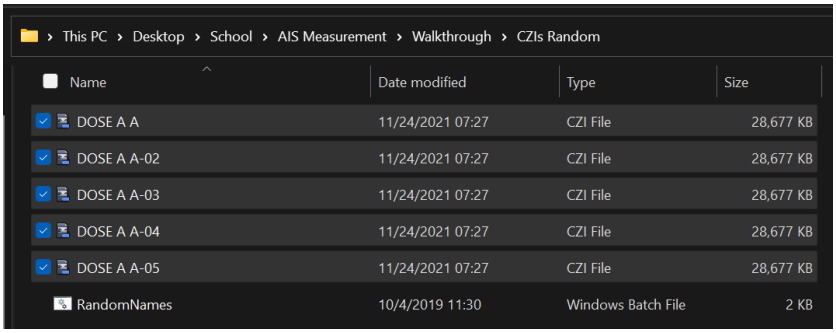
Create these folders



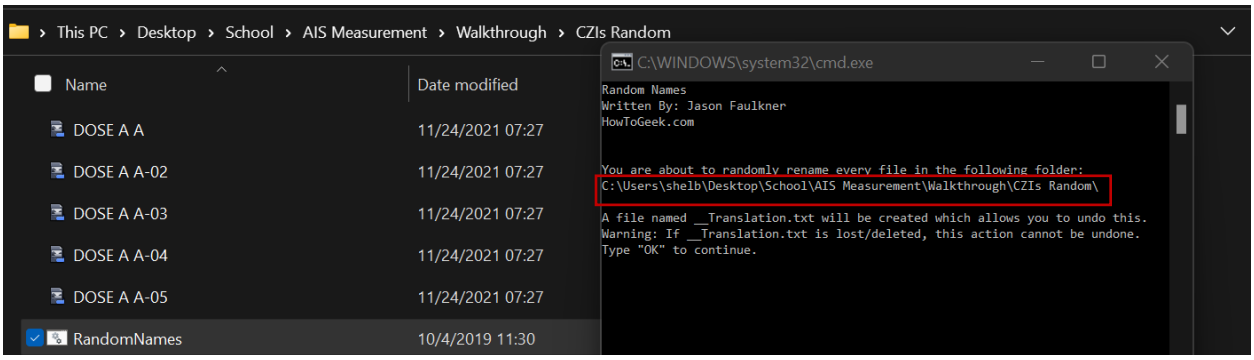

Copy “Random Names” application into “CZIs Random” folder



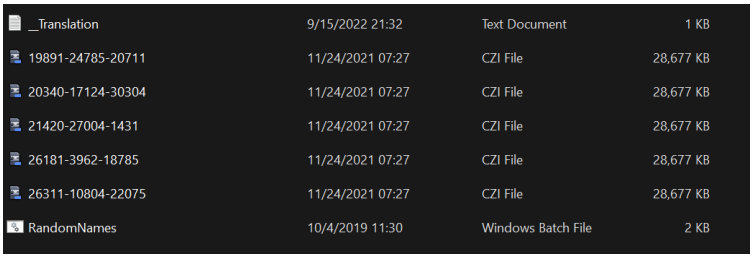

Copy all CZIs into the same folder



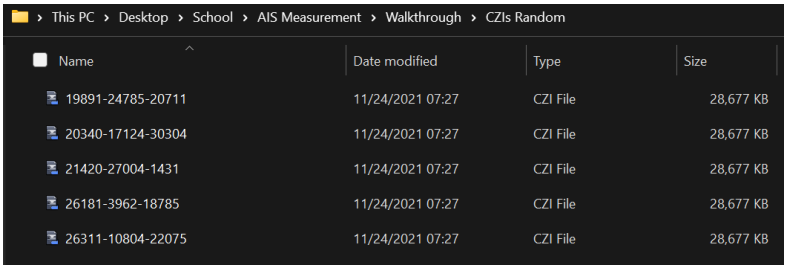

Double click Random names application in that folder to start the program.



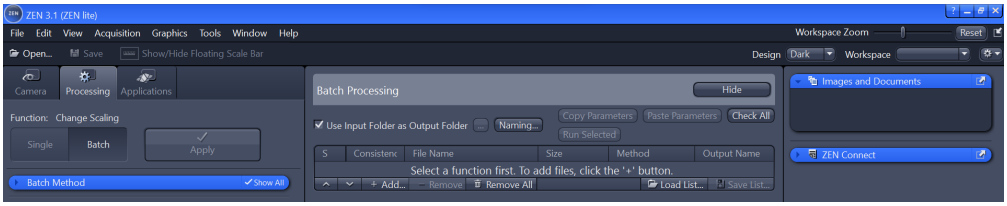

Double check that the correct folder is listed and follow prompt in window. You will now have a folder of randomized files and the Translation file with their original names.



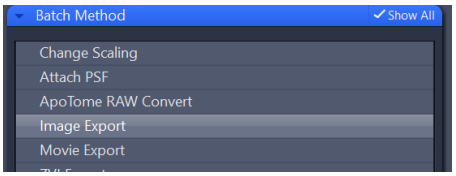

COPY translation file into original CZI folder AND email to me or put on server to access later. Then delete the translation file and random names file from the folder.



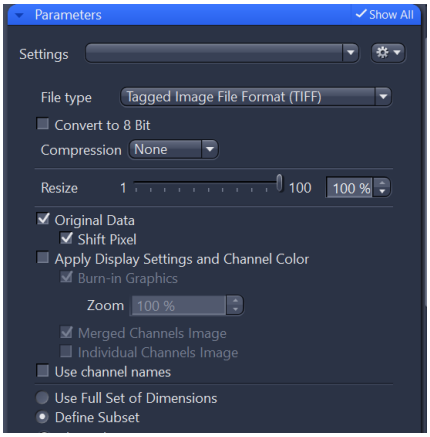

Open Zen. In the “Processing” tab select “Batch”



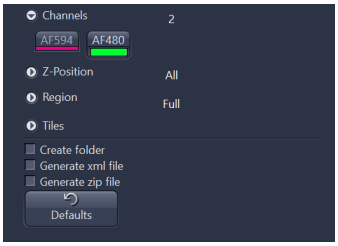

Load your randomized files into the batch processer. Then select "Image Export” as your Batch Method



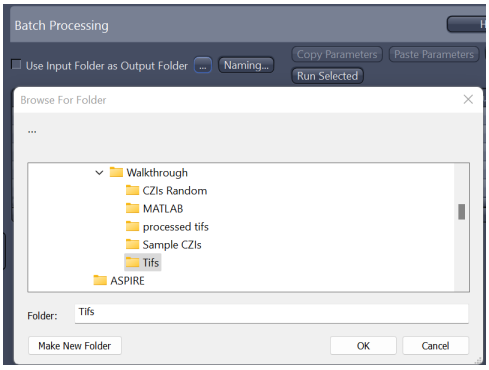

Select the following parameters exactly as they appear



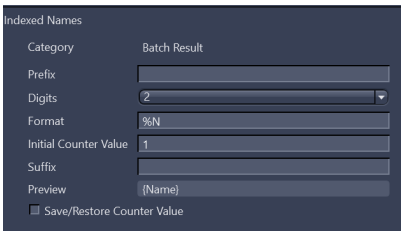

Select the channel corresponding to your AIS channel.



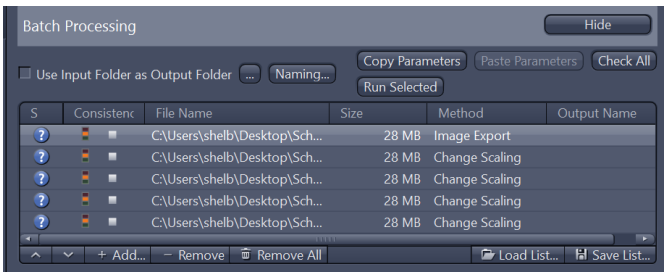

Make sure “input folder as output folder” in unchecked. Select “…” to select your output folder.



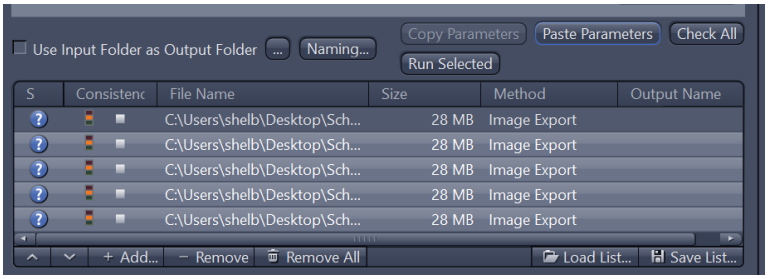

Then open “Naming…”. Your naming should look like this



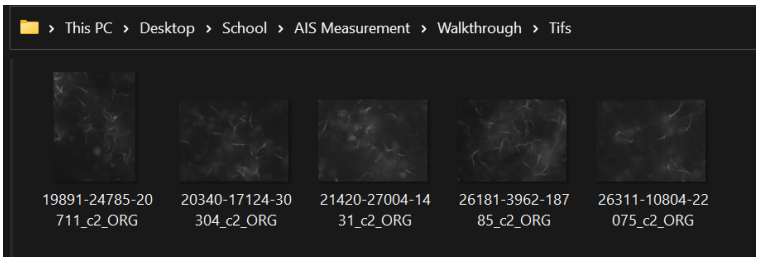

You should now see “Image Export” as the method for the first file. With this file selected, press “Copy Parameters”



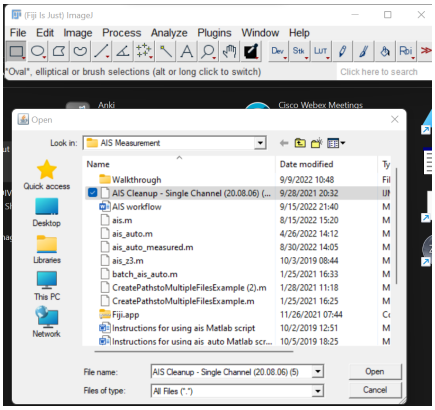

Highlight the remaining files and press “Paste Parameters”



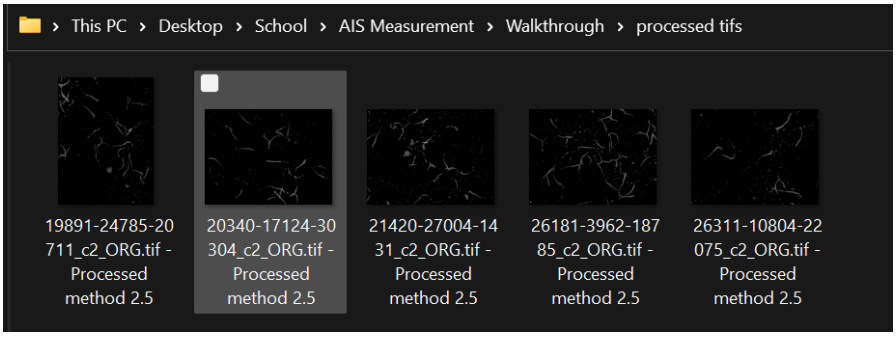

Select all files and click “Check All”. All bars under consistency should be green. If they are, press “Run Selected”. Your newly created images should now be in your “Tifs” folder. Your filenames should end in “_c2_ORG”. **If they have a different ending, make note of that.** You can now close Zen and Open ImageJ.



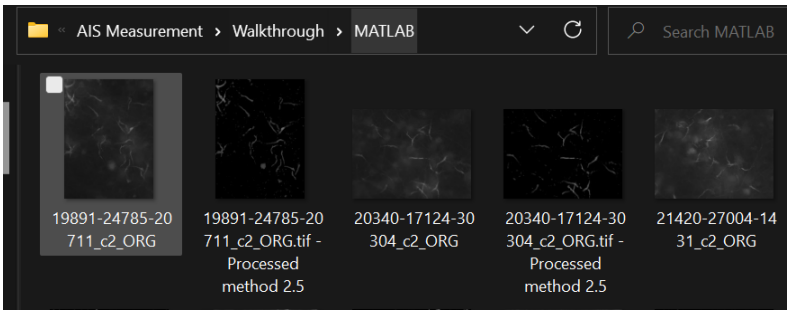

Open the AIS cleanup file



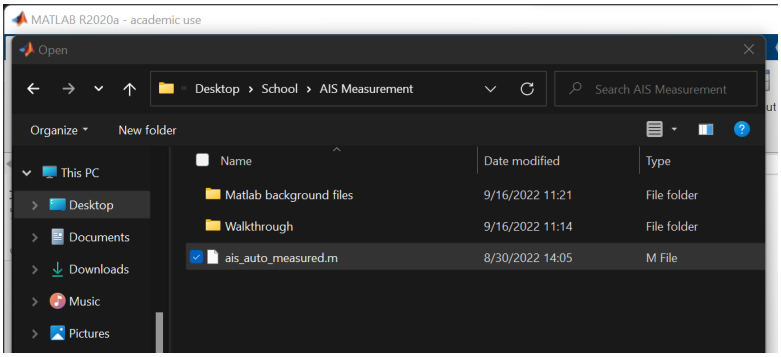

Press “Run” to begin the script. Select your “Tifs” Folder for **Images to process**, and your “Processed Tifs” folder for **Image Destination (.tif)**. You should now have background subtracted tifs in your Processed tifs folder with filenames that end in “_c2_ORG.tif - Processed method 2.5”. Again, if your filenames are different make note of that. Close ImageJ.



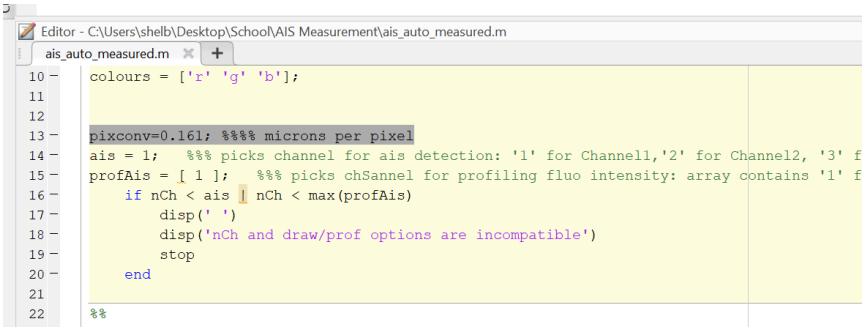

Copy the contents from your “Tifs” and “Processed Tifs” folders into your “Matlab” folder.




Now we’re ready for Matlab. Open the “ais_auto_measured” script.

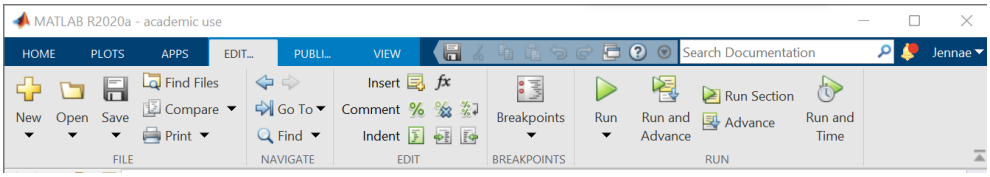



Make sure the appropriate pixel size is selected  (Listed in Zen under “info” for each CZI if you’re unsure)




Now it's time to run the script.

Under the "editor" tab, select the dropdown arrow under "Run"




Select "type code to run" and it will ask for the following parameters: (cell, threshold, smooth, chosen)

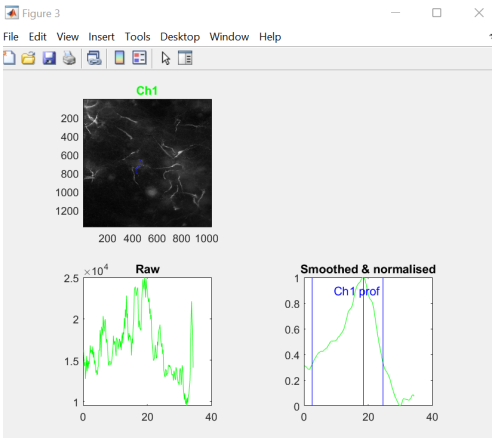
"C:\Users\w064ixa\Desktop\AIS Measurement\Walkthrough\MATLAB\19891-24785-20711_c2_ORG.tif" For cell
Right click on your first image in the MATLAB folder (Should be unprocessed tif) and select "Copy as Path" to get the file path. When you paste it, it will look like this:
"C:\Users\w064ixa\Desktop\AIS Measurement\Walkthrough\MATLAB\19891-24785-20711_c2_ORG.tif"
"C:\Users\shelb\Desktop\School\AIS Measurement\Walkthrough\MATLAB\19891-24785-20711_c2_ORG.tif"
Remove the quotations and ".tif" and replace with '
So for your code should look like this:
('C:\Users\shelb\Desktop\School\AIS Measurement\Walkthrough\MATLAB\19891-24785-20711_c2_ORG',

For threshold
Enter 0.01 (may need to play with this if not getting accurate lines and every AIS is crucial to measure)
Your code should now look like this
('C:\Users\shelb\Desktop\School\AIS Measurement\Walkthrough\MATLAB\19891-24785-20711_c2_ORG', 0.01,

Smooth and chosen will always be the same and should be 'gaussian' and 0 respectively
Your code should now look like this
('C:\Users\w064ixa\Desktop\AIS Measurement\Walkthrough\MATLAB\19891-24785-20711_c2_ORG', 0.01, 'gaussian', 0)

'C:\Users\w064ixa\Desktop\(R) MATLAB 2\17441-21240-4850_c2_ORG',0.01, 'gaussian',0

Press Enter and the processed version of your first image should appear
Follow the instructions in the command window. When AIS line is drawn, click on it and the intensity histogram will appear.
"C:\Users\w064ixa\Desktop\AIS Measurement\Walkthrough\MATLAB\20340-17124-30304_c2_ORG.tif"




The AIS length will be automatically placed in your clipboard. If the smoothed & normalized histogram looks reasonable, paste the AIS length into your Excel file. Press “Run” to repeat for all AIS in the image.

For the next image, you only need to change the file name. Press the drop-down arrow under the run button, right-click the previous code listed, and change the filename. For example:
('C:\Users\shelb\Desktop\School\AIS Measurement\Walkthrough\MATLAB\19891-24785-20711_c2_ORG', 0.01, 'gaussian', 0)
Becomes→
('C:\Users\shelb\Desktop\School\AIS Measurement\Walkthrough\MATLAB\20340-17124-30304_c2_ORG', 0.01, 'gaussian', 0)

Press Enter and repeat or all images.

